

TPG4190 Seismic data acquisition and processing

The CMP method

B. Arntsen

NTNU

Department of Geoscience and petroleum
borge.arntsen@ntnu.no

Trondheim fall 2025

Overview

- ▶ Seismic data acquisition
- ▶ CMP method 2D
- ▶ CMP method 3D
- ▶ NMO and stack
- ▶ Wide azimuth

Acquisition of seismic data

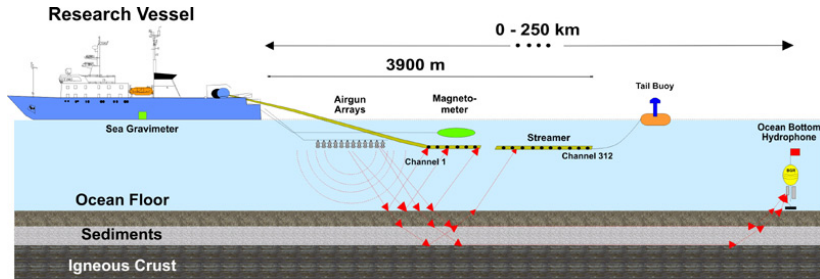
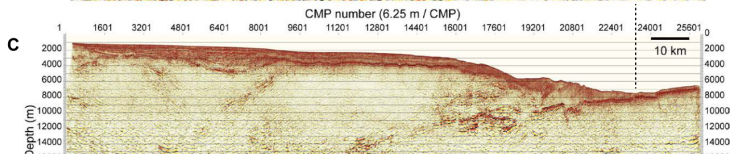
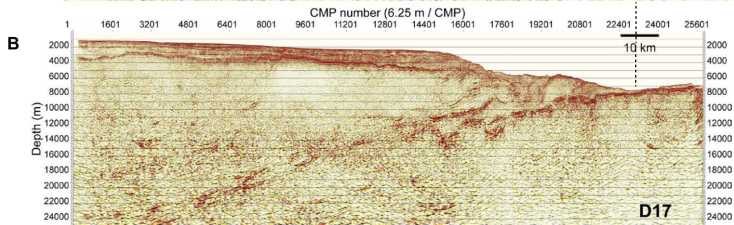
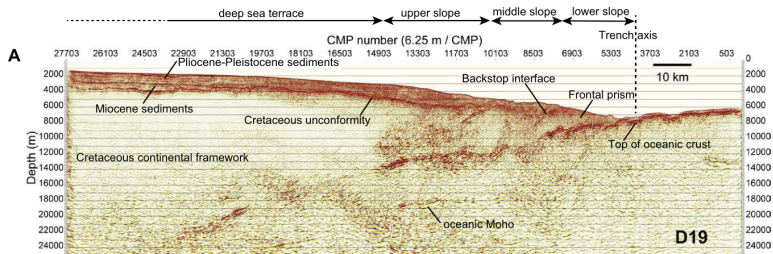
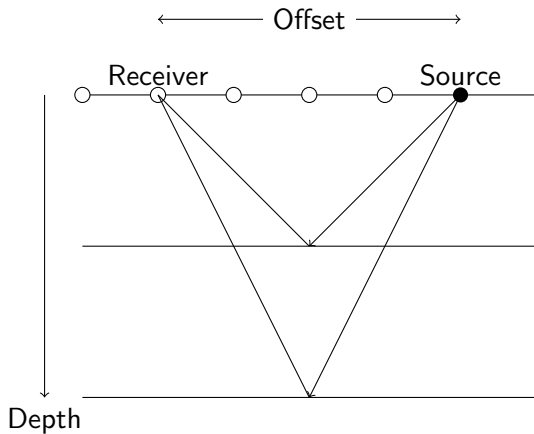


Figure: Seismic ship

Seismic data



Seismic marine data acquisition



Schematic shot record

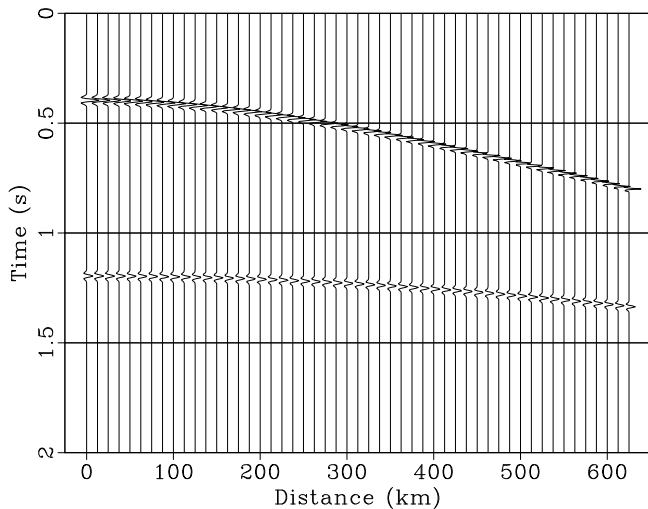


Figure: Schematic shot

Real shot record

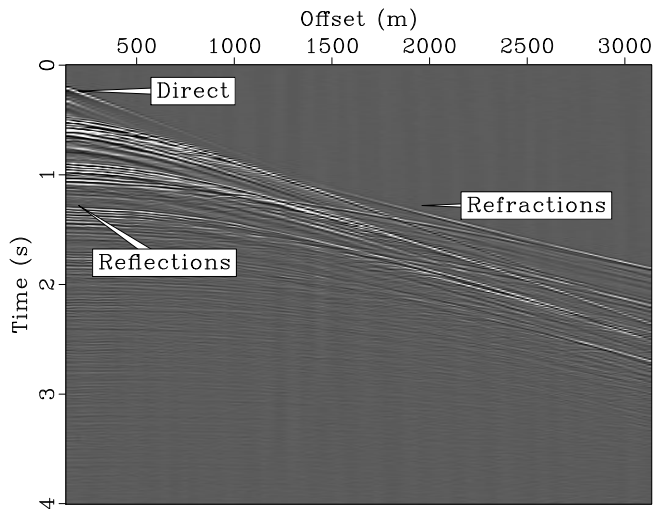
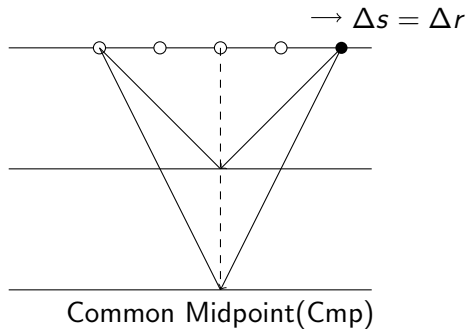
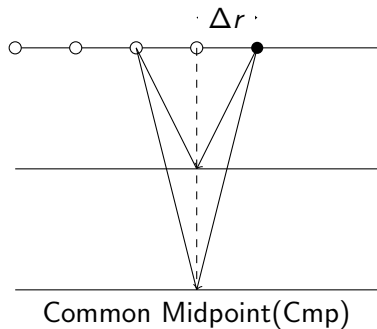
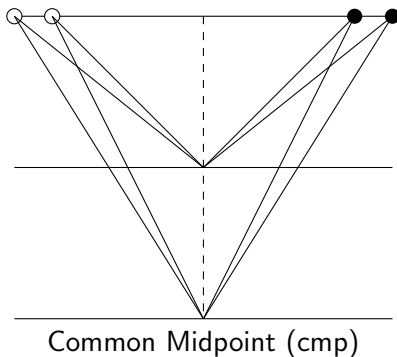


Figure: Real shot

The cmp method



Common midpoint (cmp) gather



Midpoint and Offset Coordinates

$$x_m = \frac{s + r}{2}, \quad (1)$$

$$h = \frac{s - r}{2}, \quad (2)$$

- ▶ x_m : Midpoint coordinate
- ▶ h : Offset
- ▶ s : Source coordinate
- ▶ r : Receiver coordinate

Midpoint and Offset Coordinates

Assume that we change the source and receiver coordinates a small amount δs and δr . The corresponding change in x_m and h are:

$$\delta x_m = \frac{1}{2}(\delta s + \delta r), \quad (3)$$

$$\delta h = \frac{1}{2}(\delta s - \delta r). \quad (4)$$

If we consider a single cmp then $\delta x_m = 0$

$$0 = \frac{1}{2}(\delta s + \delta r), \quad (5)$$

$$(6)$$

which implies

$$\delta s = -\delta r \quad (7)$$

Midpoint and Offset Coordinates

Inserting equation (7) into equation (4) I get

$$\delta h = \delta s = \delta r \quad (8)$$

We can now deduce the number of traces in each cmp, or the fold.
The largest (half offset) is equal to

$$h_{max} = \frac{N_r}{2} \Delta r \quad (9)$$

where

- ▶ Δr : Distance between receiver groups
- ▶ N_r : Number of receivers

The fold N_f is then

$$N_f = \frac{h_{max}}{\delta h} = \frac{N_r}{2} \frac{\Delta r}{\delta s} \quad (10)$$

Midpoint and Offset Coordinates

It remains to specify δs . The simplest assumption we can make is that $\delta s = \Delta s$, where Δs is the distance between shots. The final expression for the fold N_f is then

$$N_f = \frac{h_{max}}{\delta h} = \frac{N_r}{2} \frac{\Delta r}{\Delta s} \quad (11)$$

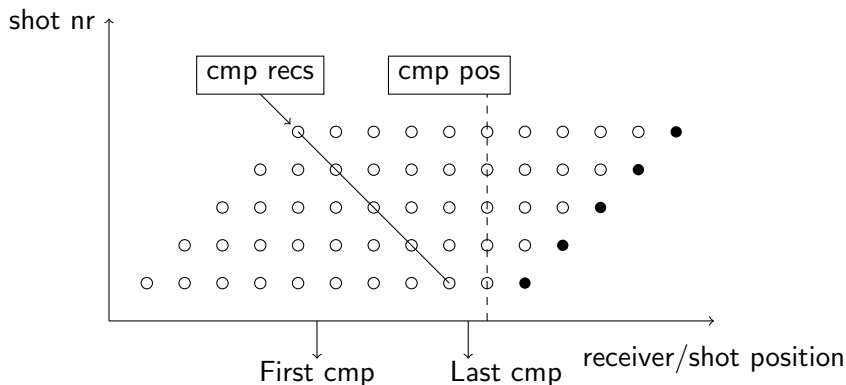
We can also figure out the distance between consecutive cmp's if we assume that the change in δr is equal to the receiver group spacing Δr and that $\Delta r < \Delta s$ and use equation (3)

$$\delta x_m = \frac{1}{2} \Delta r \quad (12)$$

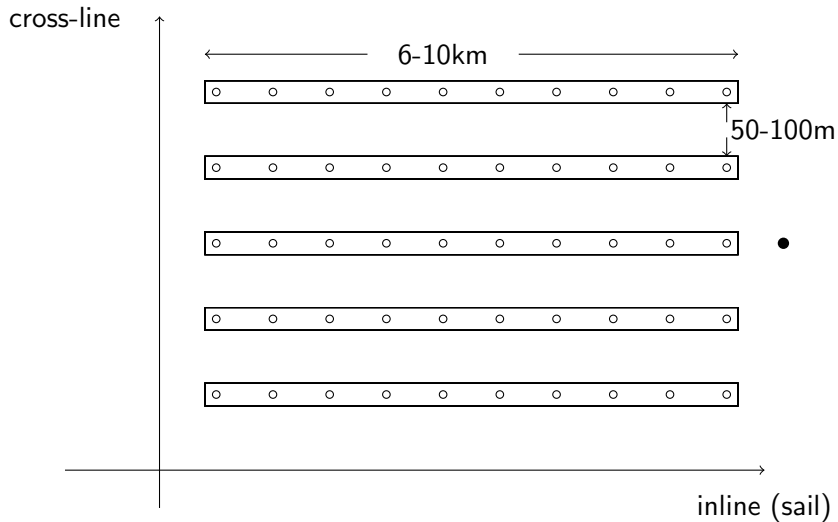
Midpoint and Offset Coordinates

Example:

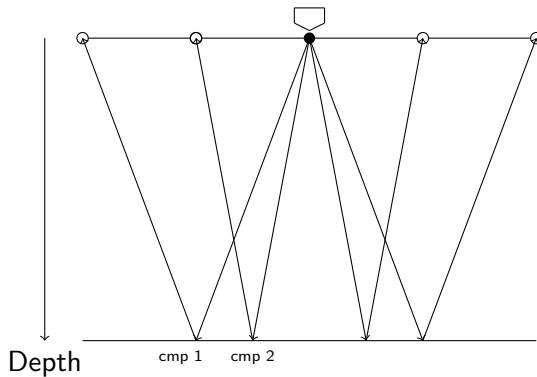
- ▶ No of receivers: 10
- ▶ No of shots: 5
- ▶ $\Delta r = \Delta s = 25.0\text{m}$
- ▶ $\Delta x_m = 12.5\text{m}$, $N_f = 10/2 = 5$



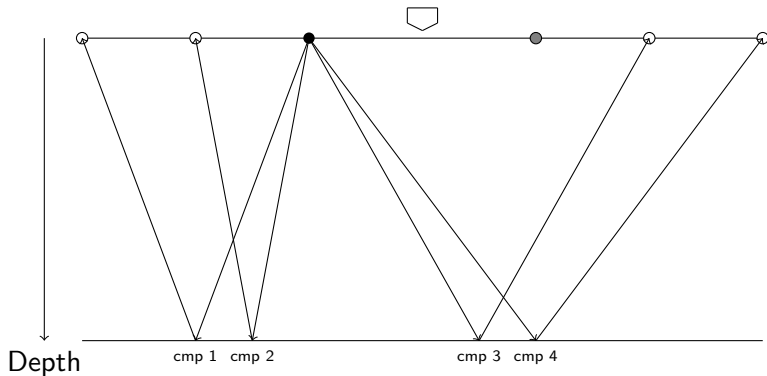
Acquisition geometry 3D



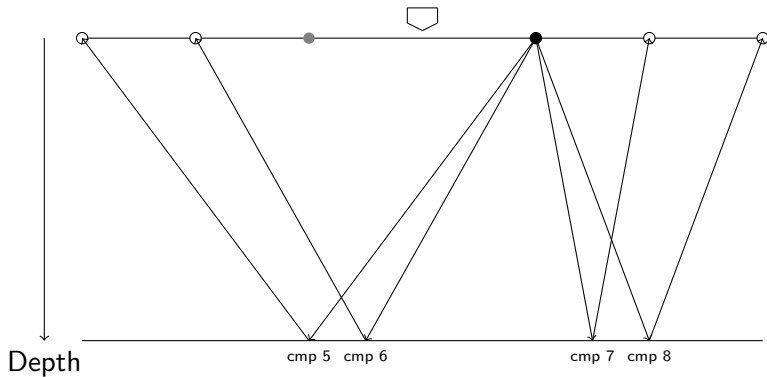
3D Seismic marine data acquisition single source



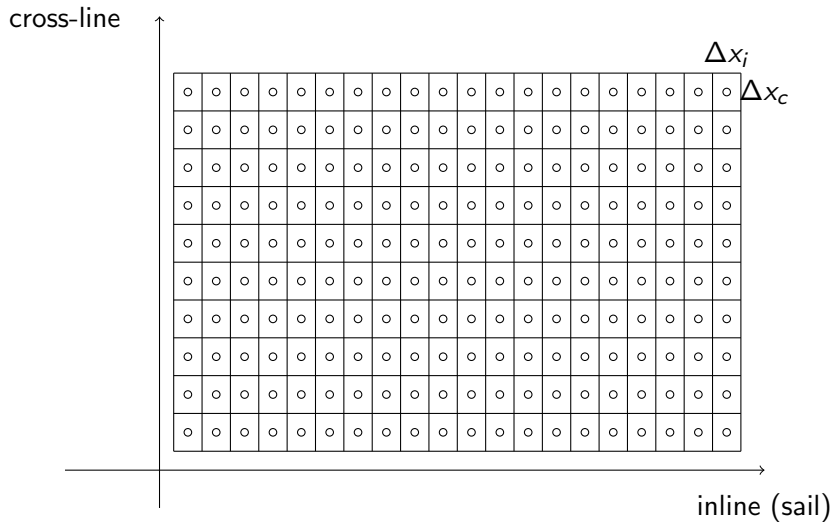
3D Seismic marine data acquisition flip-flop (left)



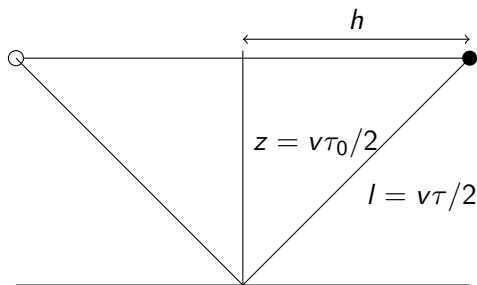
3D Seismic marine data acquisition flip-flop (right)



3D Binning



NMO and Stack



The traveltime $\tau(h)$ is:

$$l^2 = z^2 + h^2 \quad (13)$$

which gives by inserting $v\tau/2$ for l and $v\tau_0/2$ for z

$$\tau(h) = \sqrt{\tau_0^2 + 4h^2/v^2}. \quad (14)$$

NMO and Stack

Nmo-correction:

$$\Delta\tau = \tau(h) - \tau_0, \quad (15)$$

Cmp

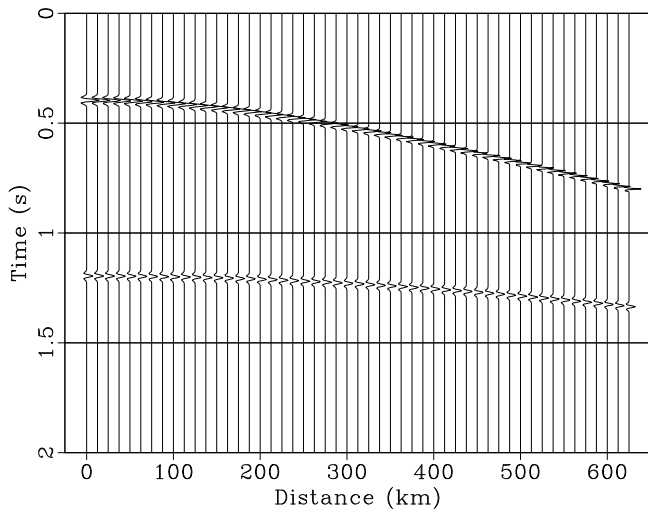


Figure:

Nmo

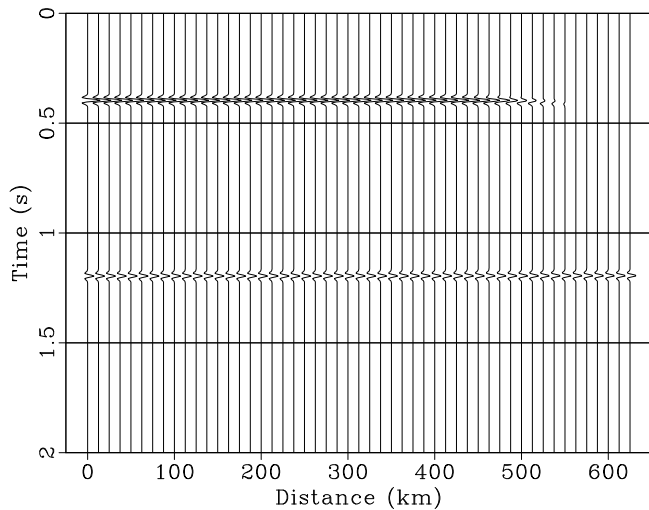
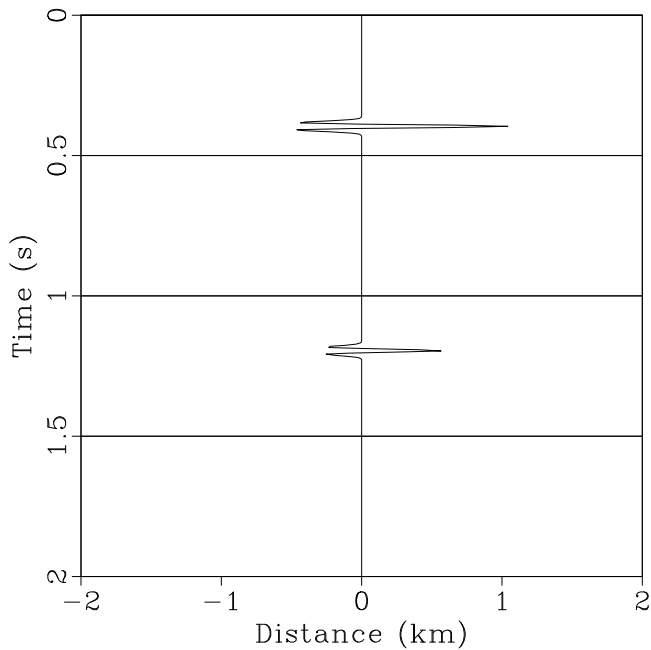


Figure: Synthetic cmp

Stack



NMO and Stack

Average velocity v_{rms} defined by

$$v_{rms}^2(t_0) = \frac{1}{t_0} \int_0^{t_0} v^2(t) dt \quad (16)$$

$v(t)$: Interval velocity. The travelttime equation (14) then becomes

$$\tau(h) = \sqrt{t_0^2 + 4h^2/v_{rms}^2(t_0)}. \quad (17)$$

Real cmp

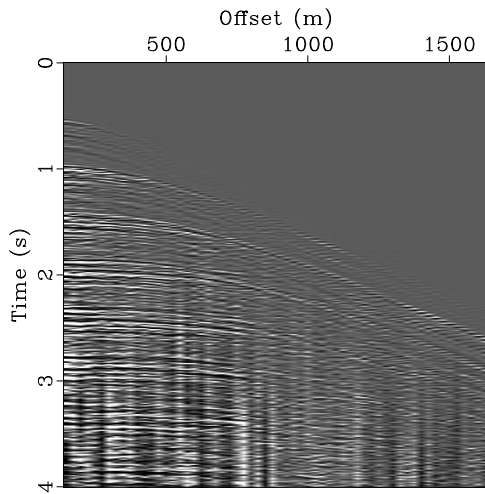


Figure: Real cmp

Real stack

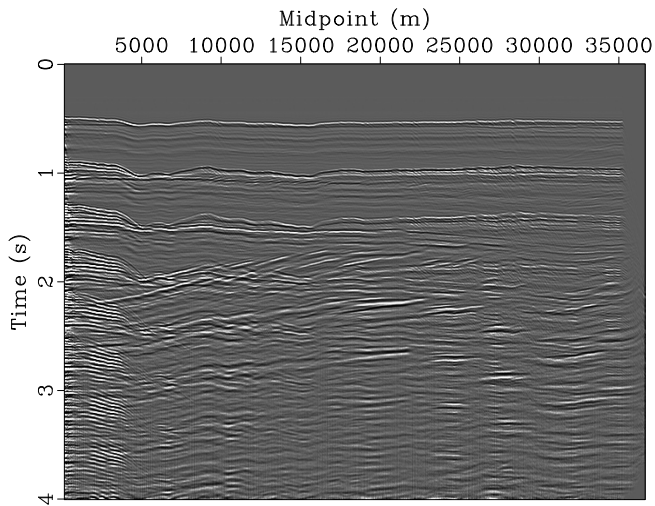


Figure: Real stack

Wide azimuth

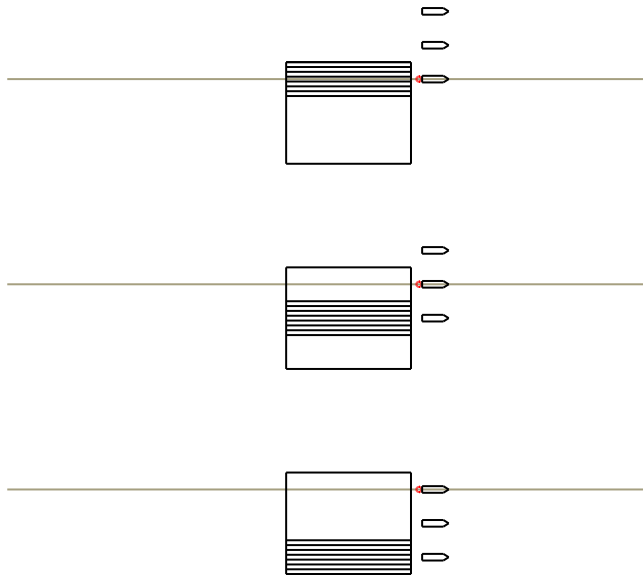


Figure: Real stack

Wide azimuth

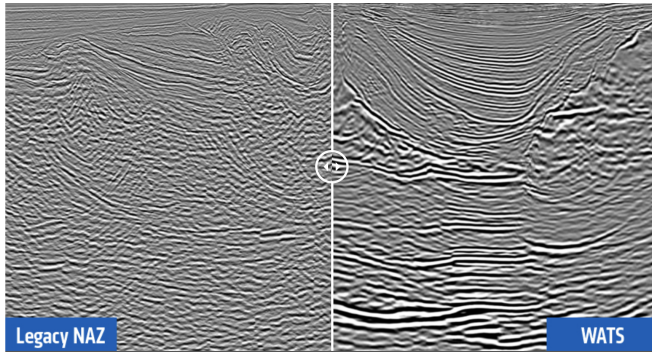


Figure: Real stack

Source: <https://www.pgs.com/publications/feature-stories/why-more-azimuths-is-a-good-thing/>